Question1: 1) Using LU decomposition, determine the matrix inverse for the following system;

$$\begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -12 \\ -20 \\ -26 \end{bmatrix}$$

2) Based on (1) find the solution?

Question2: Use multiple linear regression to fit;

x_1	0	0	1	2	0	1	2	2	1
x_2	0	2	2	4	4	6	6	2	1
y	14	21	11	12	23	23	14	6	11

Compute y(3,3).

Question3: Given these data;

x	1	2	3	5	7	8
$f(x)$	3	6	19	99	261	444

Calculate ***f(4)*** using Newton's interpolating polynomial, find f(2,3,5).